SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATION

Misty Peaks Estate

Direct-Booking Platform with Transaction-Isolated Allotment Control & AI Enquiry Engine

CLIENT
Company A (Chikkamagaluru, Karnataka)

GITHUB REPOSITORY
github.com/Imafrah/hill-resort

EVALUATION TARGET
Full-Stack Direct Booking + AI Routing

VERIFICATION STATUS
35 / 35 Tests Passing (100%)

1. Executive Summary & Problem Framing

1.1 The Business Problem

Company A operates a luxury 76-key coffee-estate resort (64 rooms + 12 villas) across 200 acres in Chikkamagaluru, Karnataka, with 140 staff and annual revenues of ₹24–28 crore. The business model suffers from two major margin drains:

1. **Online Travel Agency (OTA) Commissions (₹33.3 Lakh/year lost):**
   - 76 keys × 365 days × 62% occupancy = 17,199 room nights at ₹9,500 average tariff = ₹16.34 crore room revenue.
   - 68% sold via OTAs = 11,695 nights = ₹11.11 crore revenue, incurring 20% commission = **₹2.22 crore annually**.
   - Moving just 15% of that volume to direct direct-website bookings saves **₹33.3 lakh annually**.
2. **Event & Wedding Lead Drop-offs (₹72 Lakh/year lost):**
   - ~90 high-value wedding and corporate offsite enquiries arrive monthly across disjointed channels (GM WhatsApp, social media, general email, web form).
   - Over 30% wait >24 hours for a response, losing an estimated 3 events/month at ₹2 lakh contribution = 36 events × ₹2 lakh = **₹72 lakh annually**.
3. **Total Annual Value Captured: ₹1.05 crore / year**.

1.2 Core Solution

A direct-booking web application with:

- A streamlined **3-step booking flow** (Dates → Room → Confirm & Pay).
- **Concurrency-safe Allotment Control** via serialized transactions, guaranteeing zero overbooking against agency stock.
- A **Unified Enquiry Inbox** with a **Dual-Mode AI Router** that classifies incoming messages from WhatsApp, email, and web forms, extracts structured entities (dates, guest count, budget), assigns departmental owners, and pre-drafts replies with strict guardrails.

2. Explicitly Labeled Assumptions

As required by the assignment guidelines, all operational and domain assumptions are explicitly labeled:

- **[Assumption 1: Room & Villa Inventory Distribution]**
  The 76 keys are partitioned into 64 rooms (28 Deluxe Rooms @ ₹7,500 weekday / ₹9,750 weekend; 24 Premium Rooms @ ₹10,000 / ₹13,000; 12 Suites @ ₹18,000 / ₹23,400) and 12 villas (6 Coffee Planter's Villas @ ₹25,000 / ₹32,500; 4 Heritage Villas @ ₹32,000 / ₹41,600; 2 Presidential Villas @ ₹45,000 / ₹58,500). Total: 64 + 12 = 76 keys.
- **[Assumption 2: Web Allotment Partitioning]**
  The resort sells an admin-configurable direct allotment (`webAllotment` , seeded at ~50% of capacity). Agency allocations are managed separately; the direct site never sells outside its allotted inventory.
- **[Assumption 3: Payment Gateway Simulation]**
  Test payment is simulated with instantaneous validation, generating `pay_test_...` transaction tokens matching standard Indian payment aggregators (Razorpay/Stripe).
- **[Assumption 4: Multi-Channel Message Ingestion]**
  Direct web forms feed directly into `/api/enquiry` . Forwarded emails and WhatsApp messages are ingested through the same REST API endpoint, with an interactive testing tool in the staff UI simulating external webhooks.
- **[Assumption 5: Indian Hospitality Tax Slabs]**
  GST is applied strictly per current GST Council rules: tariffs <= ₹7,500/night incur 12% GST; tariffs > ₹7,500/night incur 18% GST.
- **[Assumption 6: DPDP Act 2023 & Foreign Form C]**
  Guest data is treated as personal data under the DPDP Act 2023. Passport details for foreign nationals are collected at physical front-desk check-in, avoiding online storage of sensitive travel documents.

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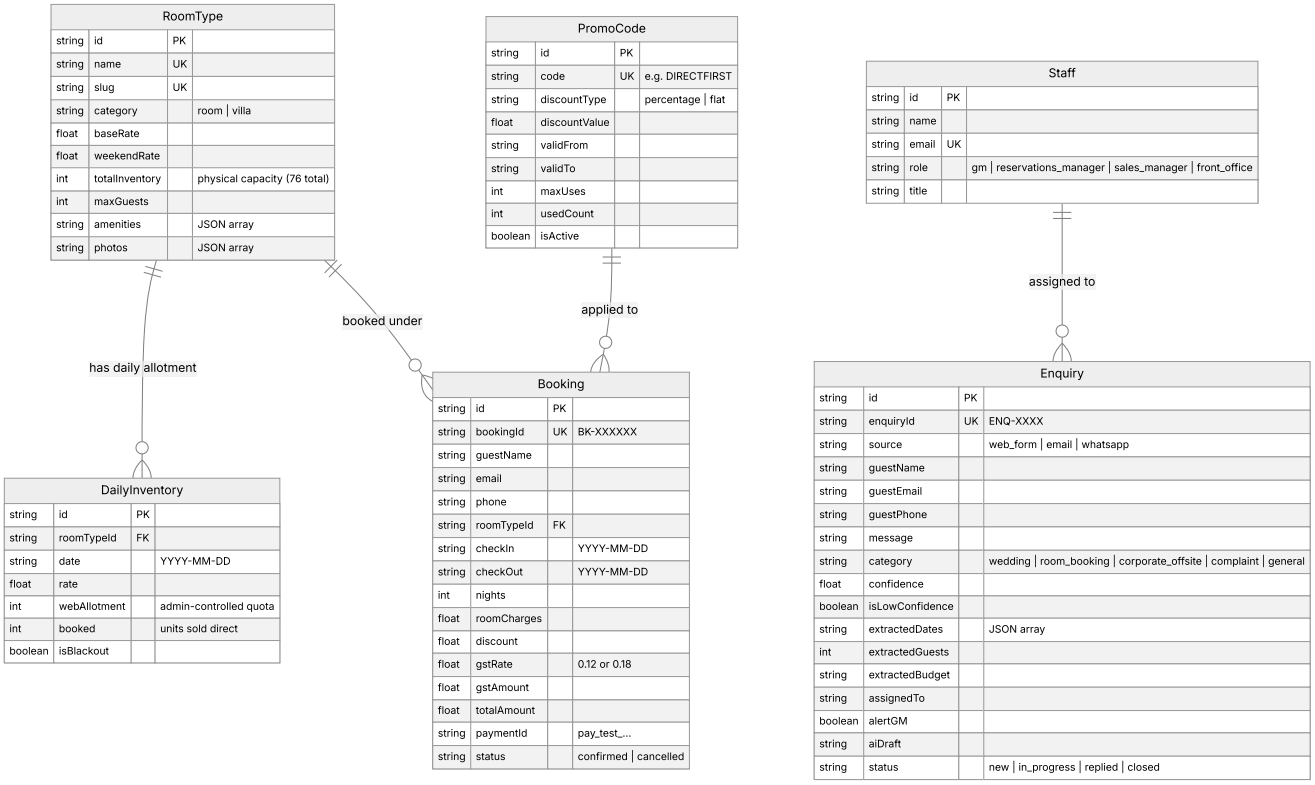
3. System Architecture & Tech Stack



3.1 Technology Decisions & Trade-Offs

LAYER	SELECTED TECH	RATIONALE & TRADE-OFFS
Framework	**Next.js 16 (App Router) + React 19**	Server Components deliver static pre-rendered HTML for instant loading on 3G/4G hill-station connections. Dynamic interactive client components are isolated to booking and calendar widgets.
Styling	**Tailwind CSS v4 + PostCSS**	Zero-runtime CSS extraction ensures high performance. Custom design tokens (`#1B3A2D` forest, `#C9A96E` gold, `#FAFAF7` cream) provide an estate-specific visual identity. Entire compiled stylesheet is under 55 KB.
Database & ORM	**Prisma 6 + SQLite (embedded) / PostgreSQL**	SQLite provides zero-configuration deployment for reviewers. Prisma's transactional isolation (`prisma.$transaction`) guarantees atomic concurrency locking. Switching to hosted PostgreSQL requires only changing the connection string.
AI Routing Engine	**Dual-Mode Engine (Local NLP + Gemini)**	**Local Heuristics Engine (Default):** Deterministic, sub-5ms keyword and entity parser running without external API keys or cost. **Gemini 1.5 Flash (Optional):** Plug-and-play LLM activated when `GEMINI_API_KEY` is provided, with automated fallback on timeout.

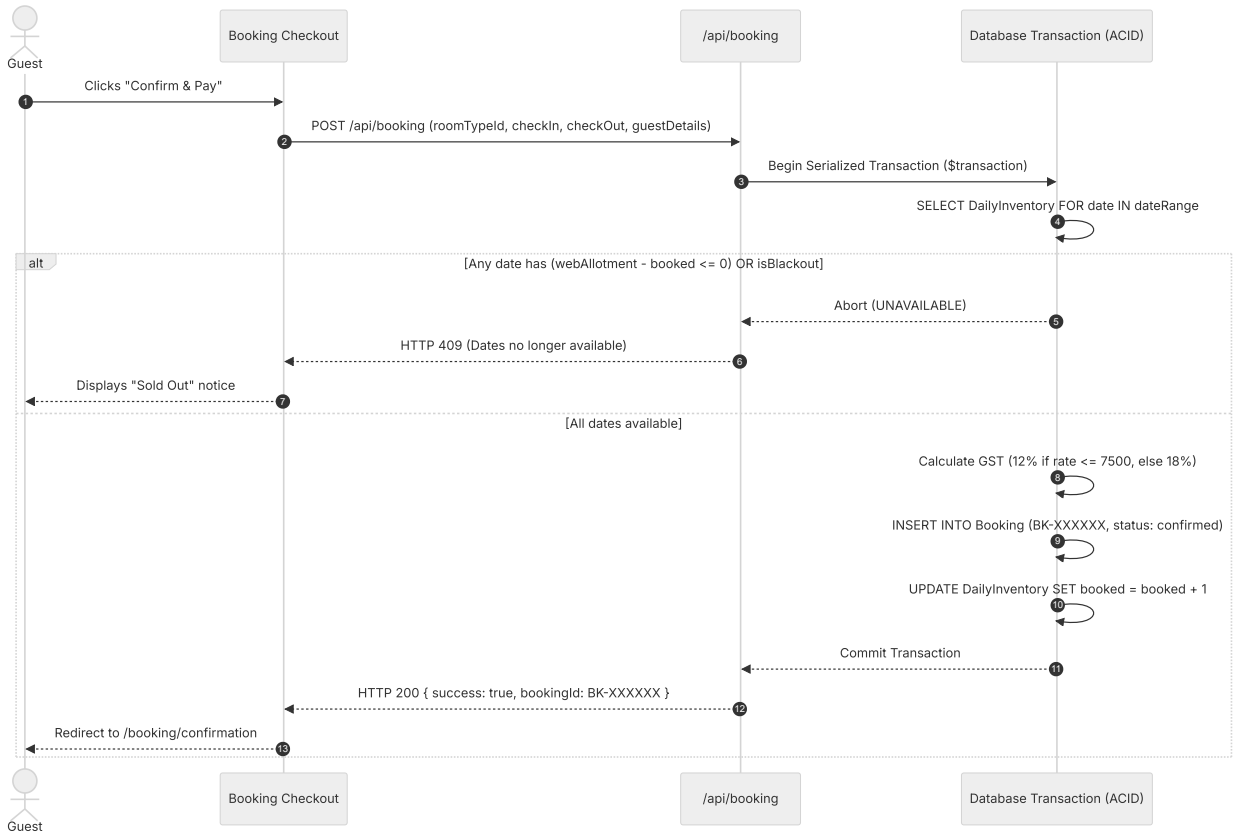
4. Data Architecture & Entity-Relationship Model



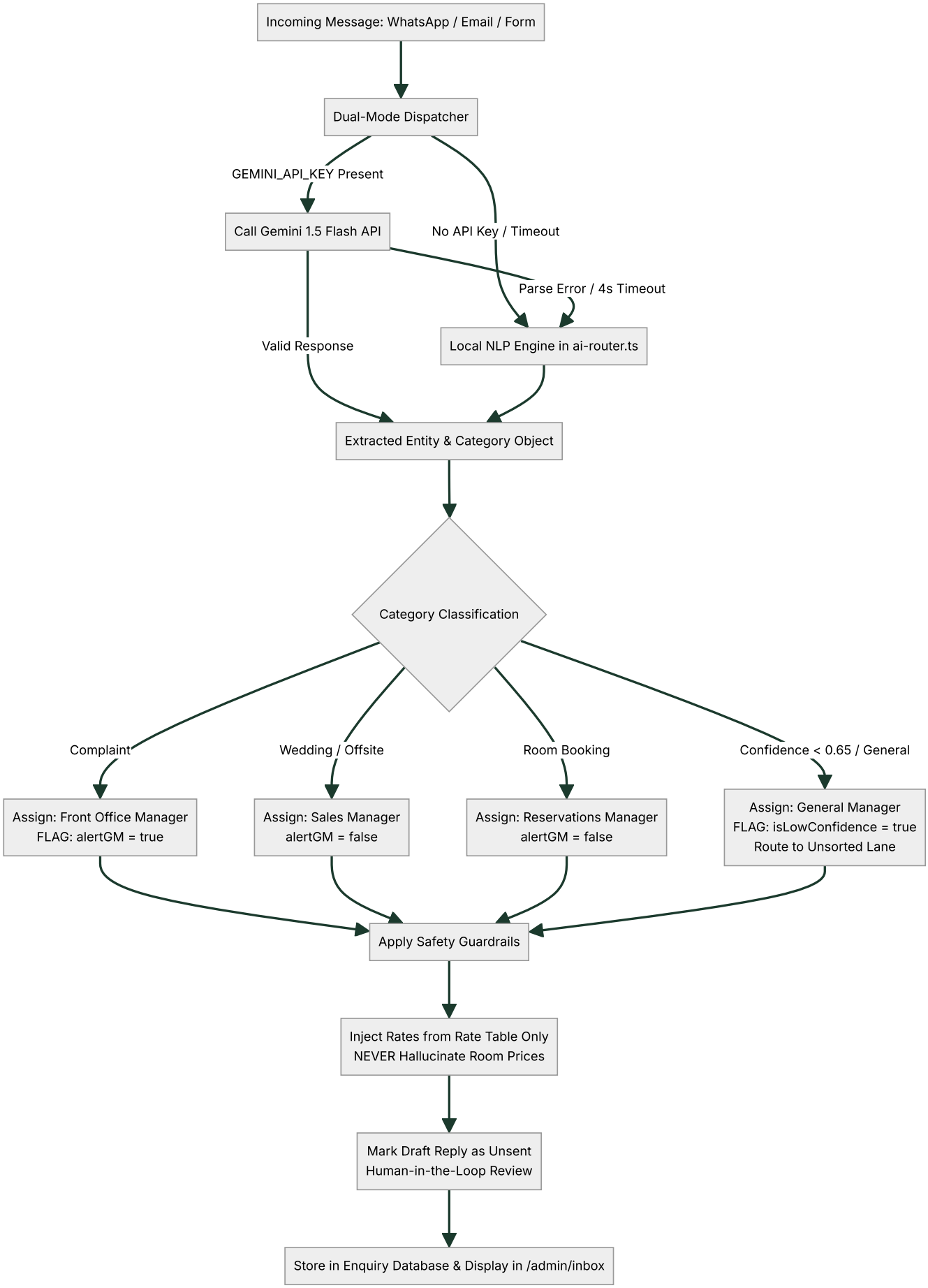
5. Core Workflows & Algorithmic Designs

5.1 Concurrency-Safe Allotment Control (Preventing Overbooking)

The central engineering constraint states: *"The site sells only an admin-set allotment, so it cannot overbook agency stock."*


Guest
Booking Checkout
/api/booking
Database Transaction (ACID)
Clicks "Confirm & Pay"
POST /api/booking (roomTypeId, checkIn, checkOut, guestDetails)
Begin Serialized Transaction ($transaction)
SELECT DailyInventory FOR date IN dateRange
alt
[Any date has (webAllotment - booked <= 0) OR isBlackout]
Abort (UNAVAILABLE)
HTTP 409 (Dates no longer available)
Displays "Sold Out" notice
[All dates available]
Calculate GST (12% if rate <= 7500, else 18%)
INSERT INTO Booking (BK-XXXXXX, status: confirmed)
UPDATE DailyInventory SET booked = booked + 1
Commit Transaction
HTTP 200 { success: true, bookingId: BK-XXXXXX }
Redirect to /booking/confirmation
Guest
Booking Checkout
/api/booking
Database Transaction (ACID)

5.2 AI Enquiry Routing Pipeline & Guardrail Logic



6. UI & Design System Decisions

The client brief mandates: *"Clean layouts, white space, a restrained palette and clear type, with nothing on screen that lacks a purpose. Design counts as much as features."*

6.1 Color Palette & Typography

- **Deep Forest Green (`#1B3A2D`):** Primary branding color symbolizing the 200-acre coffee canopy.
- **Warm White / Cream (`#FAFAF7`):** Background base providing generous white space and reducing glare on mobile screens.
- **Coffee Brown (`#6B4226`):** Secondary accent for category labels and estate heritage markers.
- **Heritage Gold (`#C9A96E`):** Call-to-action color for direct booking buttons and badges.
- **Typography:** Playfair Display (editorial serif) for headings communicates heritage and calm luxury; Inter (clean sans-serif) ensures legibility for rates, specs, and forms.

6.2 Key Screen Rationale

1. **Homepage (`/`):** Focuses on photography and high-level accommodation overview rather than dense booking widgets, immediately communicating the mountain estate experience.
2. **Room Details (`/rooms/[slug]`):** Sticky availability checker on desktop; per-night breakdowns prevent price surprise before checkout.
3. **Three-Step Booking (`/booking`):** Strictly capped at 3 steps (Dates → Room → Confirm & Pay). Progress bar at the top reassures guests of immediate completion.
4. **Structured Event Form (`/enquiry`):** Groups inputs logically into "Your Details" and "Event Details", allowing guests to specify catering, lawns, or AV needs up front.
5. **Staff Inbox (`/admin/inbox`):** Split-pane layout modeled after modern support inboxes, allowing staff to triage enquiries on the left while reviewing details, re-routing, and inspecting draft replies on the right.

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7. Scope Trade-Offs: What Was Cut & What Would Be Done Next

7.1 What Was Cut (Deliberate Out-of-Scope Choices)

To ensure the core direct-booking and AI routing slice was delivered with high quality, the following items were intentionally excluded as guided by the brief:

- **Two-Way Channel Manager Replacement:** Rather than building a fragile channel manager clone, we built an admin-set allotment partition and a daily CSV export that integrates into Company A's existing reservation software.
- **Loyalty Program & Restaurant Booking:** Kept out of scope to maintain focus on high-margin room and event revenue.
- **Online Passport Uploads:** Cut to comply with DPDP Act 2023 and local Form C regulations (checked physically at the front desk).

7.2 Next Steps (Production Roadmap)

1. **Live WhatsApp Business Webhook Integration:** Connect Meta WhatsApp Cloud API webhooks directly to `/api/enquiry` to receive and send messages in real-time.
2. **Two-Way iCal / PMS Sync:** Implement iCal feed exports and automated daily SFTP uploads for PMS synchronization.

3. **Production Payment Gateway:** Replace the test-mode mock with Razorpay live checkout supporting UPI, Credit Cards, and Net Banking.
4. **Multi-Language Support:** Add Kannada and English language toggles for regional corporate and wedding clientele.

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8. Verification & Test Evidence

8.1 Automated Test Suite

Run the test suite with:

```
npm test
```

Result: **35 tests passing, 0 failing**, covering statutory GST tiers, boundary rates, all benchmark enquiry messages, GM alert escalations, and ID generation.

8.2 Seeded Benchmark Verification

ENQUIRY ID	CHANNEL	INPUT MESSAGE	CLASSIFICATION	CONFIDENCE	ASSIGNED OWNER	GM ALERT
ENQ-4410	WhatsApp	*"Venue for my sister's wedding in Feb 2027, about 250 guests, 2 nights"*	**wedding**	92%	Sales Manager	False
ENQ-4411	Web Form	*"Is a villa free 14-16 Nov for 2 adults and a child?"*	**room_booking**	85%	Reservations Manager	False
ENQ-4412	Email	*"Team of 45 wants a January offsite with a conference hall"*	**corporate_offsite**	88%	Sales Manager	False
ENQ-4413	Web Form	*"The room was dirty and AC was not working. Worst experience. I want a refund."*	**complaint**	95%	Front Office Manager	**True (Alerted)**
ENQ-4414	Email	*"Hi, please send pictures and brochure."*	**general**	35%	General Manager	**True (Unsorted)**